Higher National Diploma in Accountancy
First Year, 2nd Semester Examination - 2021
DA 1223- Management Accounting

Instructions for Candidates:	No. of questions	: 06
Answer five **(05) questions including Question No 01**	No. of pages	: 06
Calculator is allowed	Time	: 03 hours

Question No 01

(i) State the objectives of Management Accounting? (02 Marks)

(ii) What are the differences between **Management accounting** and **Financial Accounting**? (02 Marks)

(iii) Briefly explain the **cost classification** according to the element. (02 Marks)

(iv) Define the concept of "**Material Control**". (02 Marks)

(v) What are the causes for labour turnover (02 Marks)

(vi) The following information is available in respect of component "HP001"

Minimum consumption 12,500 units — Reorder Quantity 50,000 units

Maximum consumption 25,000 units — Estimated delivery period 6 to 10 weeks

Required to calculate:

(a) Minimum Stock Level and Maximum Stock Level

(b) Re-order Level (03 Marks)

(vii) PX Company Ltd estimates the carrying cost of their stocks at 15% of unit cost and its ordering cost at Rs. 900 per order. The estimated monthly requirements are 4,000 units at a price of Rs.400 per unit.

Required calculate Economic Order Quantity (EOQ) (03 Marks)

(viii) SUN Lanka Company makes a single product and Sales price is Rs.40 per unit. Fixed cost is 480,000 and Variable cost is 24 per unit. Production and Sales units are 45,000 units

Calculate the followings.

(a) Breakeven point in units and Break-even point in sales.

(b) Margin of safety units (04 Marks)

(Total 20 Marks)

Question No 02

(a) Mr.D.J.Jhone is an employee of LUMINEX Pvt Ltd which is a paint manufacturing company. He is paid at the rate of Rs. 1200 per hour. He has to work six days a week which includes eight working hours per day. He is allowed 30 minutes to make the paint called "NE 025" and D.J.Jhone has produced 84 units of the product during a week. Differential to be applied 80% of piece rate below standard and 120% of piece rate at or above standard.

From the above particulars, you are required to;

i. Calculate the total earnings of the worker under Time rate system and Straight piece rate system
ii. Calculate total earnings under Taylor's differential piece rate system and Merrick's Multiple piece rate system. (08 Marks)

(b) SINOPEC company has three factories where they produce garments. The following information is related to Kandy factory. This factory has two production departments and two service departments for which the following information is available.

Rent and Rates	Rs.400,000	Electricity	Rs120,000
Insurance of stocks	Rs.800,000	Plant Depreciation	Rs.240,000

Additional information

	Production Departments		Service Departments	
	Cutting	Stitching	Canteen	Stores
Direct Material (Rs.)	200,000	280,000	100,000	50,000
Indirect Material (Rs.)	50,000	60,000	80,00	40,000
Indirect wages (Rs.)	100,000	80,000	75,000	125,000
Cost of Plant (Rs.)	750,000	500,000	150,000	100,000
Value of Stock (Rs.)	600,000	500,000	100,000	400,000
Floor Area (sq. ft.)	500	1200	400	400
Lighting points	120	50	10	20

Overhead cost of service departments of Canteen and Stores are apportioned as follows

Department	Cutting	Stitching	Canteen	Stores
Canteen	40%	40%		20%
Stores	50%	40%	10%	

You are required to,

(i) Prepare a statement to show apportionment of overhead cost to production and service departments.

(ii) Re-distribute overhead cost of service departments to production departments using simultaneous equation method. (12 Marks)

Question No 03

(a) Briefly explain Abnormal Loss and Abnormal Gain in process costing. (03 Marks)

(b) A manufacturing company produces a Product which is called "DRX 2". There is no opening or closing stocks and work-in-progress. The following information relates to two consecutive processes.

Particulars	Process 1	Process 2
Raw material introduced	20,000kg at Rs.40	10,000kg at Rs.15
Labour cost	2,000 Hours at Rs. 350 per hour	3,000 hours at Rs. 220 per hour
Variable production Overhead	2,000 Hours at Rs. 60 per hour	3,000 hours at Rs. 100 per hour
Fixed production cost	Rs. 205,000	Rs.431,250
Actual Output	17,500kg	25.250 units
Expected loss	5% of raw material inputs	10% of all inputs
Scrap sales value of the loss	Rs.20per kg	Rs.40 per kg

You are required to prepare the followings;

a) Process 1 and Process 2 Accounts.
b) Abnormal Loss/ Gain Account
c) Scrap sales account (17 Marks)

(Total 20 Marks)

Question No 04

(a) What are the possible causes for Material Price variance and Idle Time variance?

(04 Marks)

(b) DANKOTUWA Ltd manufactures one product, and the entire product is sold as soon as it is produced. The company operates a standard costing system and analysis of variance is made every month. The standard cost card for the product "DP T101" is as follows.

		(Rs.)
Direct Material	2kg @ Rs.500 per kg	1,000
Direct wages	2 hours @ Rs.200 per hour	400
Variable Overheads	2 hours @ Rs.400 per hour	800
Fixed Overheads	2 hours @ Rs.100 per hour	200
Standard Cost		**2,400**
Standard Profit		600
Standard Selling Price		**3,000**

Budgeted Output for the month of January 2022 was 6000 units.

Actual results for January 2022 were as follows.

- Actual production was 6,400 units
- Material purchased was 14,080kg at a cost of Rs. 6,478,800. = 460.14
- Labour hours paid for amounted to 11,520 hours at a cost of Rs. 2,534,400. = 220
- Variable overheads amounted to Rs. 5,280,000.
- Fixed overheads amounted to Rs.1, 468,000.

You are required to calculate:

(i) Direct Material Price Variance and Usage Variance.

(ii) Direct Labour Rate Variance and Efficiency Variance.

(iii) Variable Overhead Expenditure and Efficiency Variance

(iv) Fixed Overhead Expenditure and Volume Variance

(16 Marks)

(Total 20 Marks)

Question No 05

(a) The Sales Manager of ALCO & Company Ltd reports that next year he expects to sell 50,000 units of a certain product (DY 001). The standard selling price per unit is Rs. 200/.

The production manager provides his figures as follows:

Two kinds of raw materials "RM X" & "RM Y" are required for manufacturing the product "DY 001". Each unit of the product requires 2kg of RM X at Rs.10 per kg and 5kg of RM Y at Rs. 20 per kg. The estimated opening balances at the commencement of the next year are finished product 10,000 units, Raw material RM X 12,000Kg and RM Y 16,000Kg.

The estimated closing balances at the end of the next year are finished product 16,000 units, Raw material RM X 14,500Kg and RM Y 17,500Kg.

You are required to prepare,

I. Production Budget

II. Material purchase Budget (08 Marks)

(04 Luwix Company is expected to have Rs. 260,000 in its cash balance on 1st April 2023.

Prepare a cash budget for April, May, and June 2023 from the following information.

Month	Sales	Purchases	Wages	Production Overhead	Admin Expenses	Selling Expenses
March	640,000	550,000	120,000	60,000	65,000	62,000
April	720,000	500,000	80,000	50,000	50,000	37,000
May	680,000	400,000	70,000	70,000	60,000	38,000
June	920,000	650,000	100,000	40,000	70,000	44,000

Additional Information

- Sales consist 20% of cash and balance on Credit. Credit sales are realized in the month following the sales.
- All Purchases are on credit and suppliers are paid in the month following the month of purchases.
- Wages are paid instantly and expenses are paid in the next month.
- Monthly loan instalment of Rs. 20,000 has to be paid.
- Commission of 5% on Credit sales is to be paid within the month following the sales.

(12 Marks)

(Total 20 Marks)

Question No 06

(a) Briefly explain about Balance Scorecard and different perspectives of Balance Scorecard. (03 Marks)

(b) What are the key benefits of using a Balance Scorecard? (03 Marks)

(c) What are the stages involved in Activity Based Costing? (04 Marks)

(d) HIGH LIFT Company has five activity cost pools and two products. It expects to produce Product X 20,000 units and Product Y 10,000 units. Having identified its activity cost pools and the cost drivers for each cost pool, HIGH LIFT Company accumulated the following data relative to those activity cost pools and cost drivers.

Annual Overhead Data			Expected use of cost drivers per activity	Expected use of cost drivers per product	
Activity Cost Pools	Cost Drivers	Estimated Overhead (Rs.)		Product X	Product Y
Ordering and receiving	Purchase orders	400,000	5,000 orders	2000 orders	3000 orders
Machine setup	Setups	900,000	12,000 setups	5000 setups	7000 setups
Assembling	Parts	360,000	60,000 parts	36,000 parts	24,000 parts
Inspecting and testing	Tests	140,000	7,000 tests	4,000 tests	3,000 tests
Machining	Machine hours	120,000	40,000 hours	15,000 hours	25,000 hours

Using the above data, you are required to:

i. Calculate the activity-based overhead rates per cost driver.

ii. Calculate the overhead cost per unit for each product under the Activity Based Costing system (10 Marks)

(Total Marks 20)